

# PENGYU WANG

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## EDUCATION

**University of Washington, Michael G. Foster School of Business**      Seattle, WA, USA  
*Ph.D. in Operations Management*      Jan. 2023 – expected 2027  
Advisors: Prof. Shi Chen, Prof. Yong-Pin Zhou, Prof. Masha Shunko, Prof. Mingwen Yang

**University of Washington, Department of Economics**      Seattle, WA, USA  
*Ph.D. in Economics* (program; transitioned to OM)      Sept. 2020 – Dec. 2022

**University of Wisconsin–Madison, Department of Economics**      Madison, WI, USA  
*Master of Science in Economics*      Aug. 2018 – June 2020

**Western University**      London, Ontario, Canada  
*Bachelor of Arts, Honors Specialization in Economics*      Sept. 2014 – June 2018

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## RESEARCH INTERESTS

**Topics:** Sustainable operations, Innovative Platform operations, AI infrastructure operations, Global supply chain challenges, Semi-conductor industry and supply chain.

**Methods:** Stochastic optimization, Game theory, Data analysis & Causal inferences, Mean-field games and control, Reinforcement learning.

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## RESEARCH

(†: Co-first authors).

### Working Papers

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1. **Wang, P.**, Chen, S., Hou, X. (2026). *Collective ESG Investment: Mitigation, Greenwashing, and Dynamic Strategic Interactions*. Reject & Resubmit at *Manufacturing & Service Operations Management* (Apr, 2026), Resubmitted (Aug, 2026).
  - Presented at INFORMS 2025.
  - \*Scheduled presentation at INFORMS 2026. **Date:** Tuesday, November 3, 4:15 PM-5:30 PM. **Room:** Moscone South-70 (Lower Mezz).
  - We build a dynamic stochastic framework to study how heterogeneous firms make collective ESG investment decisions, including greenwashing and substantive mitigation, when their actions are strategically interdependent and interact with an ESG-aware financial market.
2. Li, Y.<sup>†</sup>, **Wang, P.**<sup>†</sup>, Zhou, Y., Shunko, M. *Can Less Be More? A Hybrid Solution to Print-on-Demand Platforms*. Reject & Resubmit at *Manufacturing & Service Operations Management* (Nov, 2025)

- Accepted by and presented at MSOM conference 2024.
  - Studies hybrid make-to-stock / make-to-order (MTS/MTO) production and assortment regulation on print-on-demand platforms. Develops analytical models and contracts to align incentives between platforms and content creators.
3. **Wang, P.<sup>†</sup>**, Zhang, J.<sup>†</sup>, Chen, S., Yang, M. (2026). *Platform-Empowered Digital Finance and Traditional Investing: Spillover Effects through Financial Literacy and Risk Tolerance*. Under Review at *Manufacturing & Service Operations Management* (Aug, 2026).
    - \*Accepted by and scheduled talk at CIST conference 2026, and DSI conference 2026.
    - This paper studies whether digital finance substitutes for traditional channels or broadens market participation. Using individual-level investment data, we conceptualize digital finance as a gateway that reduces participation frictions, facilitates experiential learning, and promotes migration into adjacent financial markets.
  4. **Wang, P.**, Zhou, Y., Shunko, M. *Marketplace Governance and Production Strategy: Decentralized Assortment and Hybrid Fulfillment in Creator Economy*. In preparation for submission to *Management Science*.
    - \* Scheduled talk at DSI 2026 conferece.
    - We study how creator portfolios are formed in digital marketplaces and the resulting production challenges for platforms. The separation between assortment creation and production decisions creates a need for marketplace governance, which platforms can address through market instruments that reshape creator portfolios and improve performance.

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## Work in Progress

5. **Wang, P.**, Chen, S. (2026) *A Two-Layer Hybrid Graph Mean-Field Game for Distributed LLM Inference Demand*.
  - Studies the offloading policy and pricing scheme in a data center network that provides AI-computing power.
6. **Wang, P.**, Chen, S. (2026) *Global Supply Chain Networks, Dynamics, and Disruption*.
  - Build a theoretical framework to uncover how global supply chain shocks penetrates through the supply network and how firms' endogenous R&D investments accelerate or contain such effects.
7. **Wang, P.**, Zhang, J., Chen, S. (2026). *Who Are the Winners? Business Opportunity Management in High-Tech Industry*.
  - Leverage a unique semi-conductor industry leading company's data to explore how such high-tech company explores business opportunity and foster future business partnership network.

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## Proceedings

8. **Wang, P.** (2017) *The Causal Relationship between Socioeconomic Status and Smoking Behavior: Evidence from Canada*. Proceedings in *Advances in Economics, Business & Management*.
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## CONFERENCES & INVITED TALKS

Decision Science Institute 2026, San Francisco, CA (scheduled)  
INFORMS Annual Meeting 2026, San Francisco, CA (scheduled)  
MOSM 2026 Annual Meeting, Boston, MA  
POMS Annual Meeting 2026, Reno, NV  
POMS-HK 2026, Shenzhen, China  
INFORMS Annual Meeting 2025, Atlanta, GA  
SICS Annual Meeting 2025, Berkeley, CA  
INFORMS Annual Meeting 2024, Seattle, WA  
INFORMS MOSM 2024 Annual Meeting 2024, Twin-City, MN  
INFORMS Annual Meeting 2023, Phoenix, AZ

## Society Services

Session chairs: INFORMS 2024, MSOM 2024, DSI 2026, INFORMS 2026

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## TEACHING EXPERIENCE

**Invited Instructor:** (Evaluations out of 5.0).

Winter 2026, IGB 250: 4.5

Summer 2026, IGB 250: 4.6

**Teaching Assistance:**

Autumn 2023, SCM 530A: 4.6; SCM 513: 4.5.

Winter 2024, BUS AN 516A: 4.2; BUS AN 516B: 4.1.

Spring 2024, OPMGT 520A: 4.3

Winter 2025, EMBA 533A: 4.5

Spring 2025, QMETH 505A: 4.1; SCM 520A: 4.2

## REFERENCES

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### **Prof. Shi Chen**

Professor of Operations Management, David and Dana Lewis Endowed Professor  
Foster School of Business, University of Washington  
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*PhD Advisor*

### **Prof. Yong-Pin Zhou**

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### **Prof. Masha Shunko**

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*PhD Advisor*

**Prof. Mingwen Yang**  
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